

Khang Thai

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Education

University of Southern California

Master of Science in Applied Data Science

Los Angeles, CA

August 2025 – May 2027

- Relevant Coursework: Deep Learning, Machine Learning, Data Management, Web Development

University of California, Los Angeles

Bachelor of Science in Statistics and Data Science

Los Angeles, CA

September 2023 – June 2025

- Relevant Coursework: Statistical Models and Data Mining, Regression, Optimization, Monte Carlo Methods, Databases

Experience

Hologic, Inc.

AI Research Associate II

San Diego, CA

September 2026 – Present

- Built multi-step LangGraph agents (retrieve, assess, answer, review) with LLM tool calling and stateful human approval gates for regulated scientific document Q&A and report drafting
- Integrated agent workflows with GitHub Copilot via the Model Context Protocol (MCP), exposing retrieval tools and a pause/resume workflow on a shared backend
- Built an evaluation harness (195 checks) for schema, data fidelity, and reproducibility, validating end-to-end report drafts generated from heterogeneous lab data exports

Hologic, Inc.

Bioinformatics Intern

San Diego, CA

June 2026 – September 2026

- Architected a hybrid RAG pipeline combining dense vector search, BM25, and cross-encoder reranking over 15,000+ chunks from 270+ controlled assay documents, doubling top-1 retrieval accuracy
- Designed a human-in-the-loop (HITL) workflow where the LLM proposes cited content, reviewers approve, and a deterministic generator fills controlled Word templates, keeping model output out of signed deliverables
- Shipped an OpenWebUI chat interface on a FastAPI backend with grounded abstention and a report-generation tool that turns raw data packages into draft reports

NeverEnding, Inc

Software Development Backend Intern

Remote

May 2024 – August 2024

- Engineered secure Python payment backends integrating the Stripe API for subscriptions and transactions
- Designed scalable Django database schemas for user accounts and digital gifting services
- Built, tested, and shipped web features across teams using Agile workflows in Jira and GitHub

Projects

Diabetes Risk Predictive Modeling

March 2025

- Built a classification pipeline on 9,500+ clinical records, achieving 97% accuracy
- Validated generalization with 5-fold cross-validation and feature importance analysis
- Visualized clinical risk factors and feature interactions to surface actionable epidemiological insights

LA Airbnb Analysis

May 2024

- Applied NLP and LDA topic modeling to extract themes and pain points from 55,000+ unstructured guest reviews
- Built logistic regression and random forest models (76% test accuracy) to identify key drivers of Superhost status
- Created an interactive Tableau dashboard to explore 45,000+ LA listings by price, room type, and beds

Technologies

AI & Machine Learning: LLMs, Hybrid RAG, Cross-Encoder Reranking, Agentic Workflows, MCP, Tool Calling, Human-in-the-Loop, Evaluation Harnesses, NLP

Frameworks & Libraries: LangGraph, LangChain, OpenWebUI, FastAPI, Pydantic, PyTorch, scikit-learn, Django

Databases & Vector Search: Qdrant, PostgreSQL, MySQL

Languages, Tools & Platforms: Python, R, SQL, Java, JavaScript, Git, Jira, Linux/Bash, Tableau, Docker